



Community

Codetopia Mentorship Program - 2026

Git & GitHub



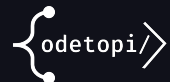
Session 2 :

The Basic Git Workflow



What we will cover today

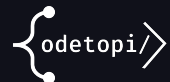
- Quick review of the 3 Git states
- Initialising a repository with `git init`
- The basic loop: edit → stage → commit
- Writing good commit messages (Conventional Commits)
- Viewing history with `git log`
- Session Project: Commit History Journal





Quick Review From Previous Session

- What are the 3 states?
- What is VCS and Version Control?
- Difference between CLI & GUI?
- How to Install and Configure your Identity



The Basic Git Workflow

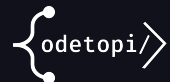


The 3 States

- Working Directory : your files as you edit them
- Staging Area : selected changes ready to commit
- Repository : permanent commit history

Today we use all 3 states for the first time:

- We will initialise a repo, stage files, commit them, and read the history.





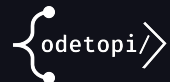
Initialising a Repository

Create your first repo

```
mkdir my-project  
cd my-project  
git init
```

What git init does

- Creates a hidden `.git` folder, this is the repository
- Your working directory is now tracked by Git
- Nothing is committed yet, the repo is empty



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Staging & Committing

The basic loop

1. Create or edit a file

```
echo 'Hello Git' > hello.txt
```

2. Check the status

```
git status
```

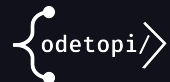
3. Stage the file

```
git add hello.txt      # stage one file
```

```
git add .              # stage all changes
```

4. Commit with a message

```
git commit -m "feat: add hello.txt"
```



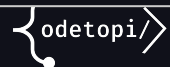


Writing Good Commit Message

The Conventional Commits format

“type: short description (under 72 chars)”

Type	When to use it
feat:	Adding a new feature
fix:	Correcting a bug
docs:	Changes to documentation or notes
chore:	Setup, config, or maintenance tasks
refactor:	Restructuring code without changing behaviour





Writing Good Commit Message

Bad vs Good Commit Message

Bad	Good
fix	fix: correct typo in recipe title
stuff	feat: add ingredient list to pasta recipe
aaa	docs: add setup instructions to README





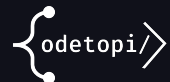
View History With Git Log

Essential log commands

```
git log                # full history
git log --oneline       # compact, one line per
commit
git log --oneline --graph    # visualise branches
git log --oneline --all --graph # include all branches
```

Reading a commit entry

- Commit hash : unique ID for this snapshot (e.g. a3f9c2d)
- Author : who made this commit
- Date : when it was committed
- Message : what changed and why





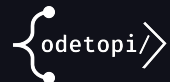
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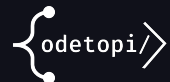
Session Project

Commit History Journal

Session Project

Your deliverables

1. Initialise a new repo for a topic of your choice (recipes, movies, travel, etc.)
2. Create 3 separate files(.md files) and commit each one with a descriptive message
3. Edit one file, stage it, run `git diff --staged`, then commit. Take a screenshot after doing this.
4. Run `git log --oneline` to show at least 5 commits screenshot it
5. Write a bad commit, then use `git commit --amend` to fix the message

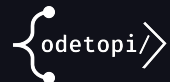




Create your Project Folder

Your home for the next 8 weeks

You'll keep all your work for the program in one folder on your computer every week's project, every screenshot, every capstone. One folder, eight weeks, everything in one place.



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Create your Project Folder

Create it now

```
# pick a spot on your computer
cd ~/Documents

# create the folder
mkdir codetopia-mentorship-program
cd codetopia-mentorship-program

# track it with Git
git init
```



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Create your Project Folder

Create it now

```
codetopia-mentorship-program/  
├─ week-1-foundation/  
│   ├─ session-2/slides  
│   │   └─ screenshots/      ← today's work goes here  
│   └─ capstone/  
│       └─ screenshots/  
├─ week-2-core-workflow/  
└─ ... through week-8
```



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Learning Achievement

By the end of this project you should have been able to initialise a repo, stage and commit changes, inspect history, and amend a mistake, all from memory, without looking at notes.



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References & Further Reading

Resources

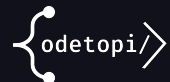
Text Resource

- The Odin Project: Git Basics
[Git Basics | The Odin Project](#)

Video Resource

Net Ninja Git Tutorial #4

[Git Tutorial for Beginners: Command-Line Fundamentals](#)



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